TECHNICAL REFERENCE & CURRICULUM GUIDE

Module 4: Model Scaling, Parameters & MoE

Module 4 examines the mechanics of model sizing, scaling laws, and the shift from dense transformers to Mixture-of-Experts (MoE) architectures. Every parameter dimension, mathematical scaling law, and routing algorithm is broken down below.

4.1 What '7B / 70B / 405B Parameters' Actually Means

When an AI model is described as a "7B", "70B", or "405B" parameter model, it refers to the **total count of scalar floating-point weights (W)** stored across all weight matrices in the network, updated via backpropagation.

1. Where Do Parameters Live in a Transformer?

Almost 90%+ of a Transformer's parameters live in two primary components:

- **Self-Attention Projections:** W_Q, W_K, $W_V \in \mathbb{R}^{d_{model} \times d_k}$ and $W_O \in \mathbb{R}^{(h \cdot d_v) \times d_{model}}$.
- **Feed-Forward Networks (FFN / MLP):** Maps hidden states up to $4 \times d_{model}$ and back down. In SwiGLU architectures, the FFN contains **three** separate weight matrices (W_{gate}, W_{up}, W_{down}).

2. Deriving Parameter Count from Architectural Dimensions

For a model with $d_{model} = 4096$, $n_q = 32$, $n_{kv} = 8$, $d_{head} = 128$, $L = 32$ layers, and vocabulary size $V = 128{,}000$:

- **Attention Projections per layer:** Q (4096×4096), K (4096×1024), V (4096×1024), O (4096×4096) $\approx 41.9 \text{M}$ parameters.
- **SwiGLU FFN per layer:** Intermediate dimension $d_{ff} \approx 11{,}008$ across three matrices $\approx 135.3 \text{M}$ parameters.
- **Total Backbone & Embeddings:** Across 32 layers plus embedding table ($128{,}000 \times 4096 \approx 524 \text{M}$), totaling $\approx 7 \text{B}$ parameters (**7B Model**).

4.2 Parameter Count vs. Capability — Why More Isn't Always Better

- **The Data Quality Trap:** A 405B model trained on low-quality web text underperforms a curated 8B model trained on 15T+ high-density tokens. Without data saturation, overparameterization leads to memorization rather than generalization.
- **Inference Economics & SLMs:** Serving a 70B model requires massive multi-GPU setups. In 2026, 1B–8B **Small Language Models (SLMs)** trained on massive token budgets match or exceed older 70B dense models while running locally at hundreds of tokens per second.

4.3 Scaling Laws: Kaplan vs. Chinchilla

- **Kaplan Scaling Laws (OpenAI, 2020):** Posited that parameters (N) and dataset size (D) scale independently, recommending adding $7\times$ more parameters for every $1.4\times$ token increase —leading to undertrained models.
- **Chinchilla Optimal Scaling Laws (DeepMind, 2022):** Demonstrated that N and D must scale in equal proportion ($D \approx 20 \times N$).
- **The 2026 Over-Training Reality:** Modern training deliberately violates Chinchilla optimality by **over-training models far past the 20:1 ratio** (e.g., 200T tokens for an 8B model) because inference is perpetual and smaller over-trained models drastically slash serving VRAM and compute overhead.

4.4 Mixture-of-Experts (MoE)

- **Core Concept:** Replaces dense FFN layers with a **Router Network** and E parallel expert FFNs. Incoming tokens are dispatched only to the **top-k experts** (e.g., $k=2$ out of 8 or 64).
- **Active vs. Total Parameters:** Decouples total capacity from inference speed. Mixtral 8x7B has ~47B total parameters but activates only ~13B per token. DeepSeek-V3 has 671B total parameters but activates only 37B active parameters per token.
- **Routing & Load Balancing:** Unconstrained routers collapse by routing 90% of traffic to 1-2 experts. Modern architectures use **Load Balancing Loss** or **Auxiliary-Loss-Free Load Balancing** to ensure uniform expert utilization.